

The Compact Topology of Quantum Representation Learning: A Fugue on Six Themes

A Formally Verified Account

34 Lean modules · 572 declarations · 0 `sorry` · 6 verification tools

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“A mathematical monograph should be like a compact topological space: every finite subcover covers all the ideas. Within any three pages, the reader should encounter all six themes—Jensen, collapse, quantum, geometry, M_2 , gradients—at whatever depth the exposition has reached.”

Six recurring themes. This paper is organized as a fugue on six themes that appear in every movement, at increasing depth:

- J** Jensen’s inequality on spectra
- C** Collapse and the trivial minimizer
- Q** Quantum feature maps and barren plateaus
- G** Geometry: fibre bundles, curvature, Lie algebras
- M** The second moment operator M_2
- ∇** Gradient variance and trainability

The themes are not independent. They form a complete graph: every pair is connected by a theorem. The paper’s four movements traverse this graph at increasing resolution.

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1 Movement I: The Energy and Its Pathology

1.1 The energy

Consider a dataset of images. A generative model predicts pixels—but this is wasteful. Most information in an image is irrelevant to understanding what it contains. LeCun (2022) proposed instead to predict *representations*: map each input x to an abstract vector $\phi(x)$ capturing semantic content, then predict in this abstract space.

Definition 1.1 (JEPA Energy). *Let $\phi : X \rightarrow Z$ be an encoder mapping inputs to a Hilbert space Z , and let $V : Z \rightarrow Z$ be a predictor. The **JEPA energy** is*

$$E(x, y) = \|\phi(y) - V(\phi(x))\|^2.$$

Three immediate consequences, each formally verified:

1. $E(x, y) \geq 0$ for all x, y (squared distance)
2. $E(x, y) = 0 \Leftrightarrow \phi(y) = V(\phi(x))$ (perfect prediction)
3. When $V = \text{id}$: $E(x, y) = \|\phi(y) - \phi(x)\|^2$ (distance between representations)

Lean: `QJEPa.qjepaEnergy_nonneg`, `qjepaEnergy_eq_zero_iff`, `qjepa_reduces_to_jepa`.

First appearance of all six themes. Before diving into proofs, we state the entire landscape—every theme that will recur throughout the paper:

- J Jensen.** The trivial minimizer exists because of a spectral principle—Jensen’s inequality—that characterizes when all eigenvalues coincide. We will make this precise.
- C Collapse.** Setting $\phi(x) = c$ for all x gives $E \equiv 0$. The global optimum is the catastrophe. No gradient-based optimizer can avoid it without intervention.
- Q Quantum.** The same energy works in $\mathbb{C}^{2^n}$. But exponential dimension makes the pathology exponentially worse: not just the loss value but its gradient vanishes.
- G Geometry.** The energy measures the failure of parallel transport—it *is* curvature, as we shall see. Zero energy = flat connection = collapsed encoder.
- M M_2 .** The concentration of kernel values is governed by a single operator M_2 whose spectrum on $\text{Sym}^2 \oplus \Lambda^2$ we will compute exactly.
- ∇ Gradient.** Not only the loss but its gradient vanishes in the barren plateau regime. The landscape is flat everywhere—not just at the minimum.

The reader now holds the complete map. Every subsequent page fills in details that connect to all six themes simultaneously.

1.2 The trivial minimizer and Jensen’s principle

Set $\phi(x) = c$ for all x , with $V = \text{id}$. Then $E(x, y) = \|c - c\|^2 = 0$ for every pair. The energy is minimized globally, perfectly, for free. No learning occurs.

This is not a pathology that requires exotic constructions to encounter. It is the *global optimum* of the energy functional. Any gradient-based optimizer, left to its own devices, will find it. The trivial solution is not a local minimum that might be escaped with a clever initialization; it is the bottom of every valley.

To see the depth of the problem, consider the kernel matrix $K_{ij} = \langle \phi(x_i), \phi(x_j) \rangle$ induced by the encoder on a training set $\{x_1, \dots, x_N\}$. If ϕ is constant, then $K_{ij} = \|c\|^2$ for all i, j —every entry is the same. The kernel has rank 1: all its eigenvalues are zero except one. The encoder has compressed an N -point dataset into a single point.

Now consider the opposite extreme: an encoder that makes all kernel eigenvalues equal (an isotropic kernel). This means $K_{ij} = \delta_{ij}$ —every point is orthogonal to every other point. The encoder has “spread out” the data maximally, but it has also destroyed all structure: nearby inputs are no farther apart than distant ones. This too is useless.

Both extremes — total compression and total dispersion — are pathologies, and both are characterized by the same mathematical condition: *spectral degeneracy* of the kernel. Why is this the optimum, not merely a local minimum? The answer is Jensen’s inequality applied to the kernel spectrum.

Theorem 1.2 (Jensen on Spectra). *For any convex function $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ and non-negative spectrum $\lambda_1, \dots, \lambda_K$:*

$$f\left(\frac{\sum_i \lambda_i}{K}\right) \leq \frac{\sum_i f(\lambda_i)}{K},$$

with equality if and only if all λ_i are equal.

Lean: `Spectrum.jensen`, using Mathlib’s `ConvexOn.map_sum_le`.

The power of Theorem 1.2 lies in choosing different convex probes f :

- $f(x) = x^2$: the **QM-AM inequality** ($\sum \lambda_i^2 / K \leq \sum \lambda_i^2$). Governs kernel risk and purity.
- $f(x) = 1/x$: the **AM-HM inequality**. Governs estimator variance.
- $f(x) = -\log x$: the **entropy bound**. Governs information content.

In every case, **equality holds iff the spectrum is isotropic** (all eigenvalues equal). Isotropy is the spectral shape that simultaneously minimizes all convex functionals. This is why VICReg sets $\Sigma = I$, why SIGReg drives embeddings toward a Gaussian, and why the maximally mixed quantum state $\rho = I/d$ achieves maximum entropy.

Theorem 1.3 (Collapse Equivalence). *The following are equivalent for an encoder $\phi : X \rightarrow Z$:*

- (i) ϕ is collapsed: $\exists c, \forall x, \phi(x) = c$.
- (ii) Zero energy for identity predictor: $\forall x, y, E(x, y) = 0$.
- (iii) All encodings are equal: $\forall x, y, \phi(x) = \phi(y)$.

Lean: `CollapseEquivalence.collapsed_iff_zero_E`.

Through Jensen’s lens, collapse has a spectral characterization. Let $K_{ij} = \langle \phi(x_i), \phi(x_j) \rangle$ be the kernel matrix.

Theorem 1.4 (Spectral Characterization of Collapse). *The encoder is collapsed iff $\sum \lambda_i^2 = (\sum \lambda_i)^2 / K$. Equivalently, collapse is Jensen equality for $f(x) = x^2$.*

Geometrically [G], Jensen’s inequality is a curvature comparison theorem. A convex function f has positive second derivative (positive curvature). The inequality $f(\bar{\lambda}) \leq \overline{f(\lambda)}$ says: the average of a curved function exceeds the function of the average, because “curving upward” means the function lies above its chords. **Equality means zero curvature**: when all λ_i are equal, the function is evaluated at a single point, and the chord degenerates to a point. No curvature is “felt.” This geometric interpretation recurs at every level: **isotropy = flatness**, and flatness is simultaneously the optimum (for embeddings) and the catastrophe (for quantum circuits [Q]).

Lean: `GeometricUnification.jensen_curvature_comparison`, `jensen_flat_iff_isotropic`.

So **collapse = isotropy = Jensen equality = zero curvature**. The same condition appears under different names:

Field	Name for isotropy
Representation learning [C]	Collapse
Quantum mechanics [Q]	Maximally mixed state (purity = $1/d$)
Statistics [J]	Maximum entropy
Geometry [G]	Zero curvature (flat connection)
Spectral theory [M]	Degenerate spectrum of M_2
Optimization [V]	Vanishing gradient (flat landscape)

Every row of this table is a theorem, not an analogy. Six themes, one mathematical condition.

1.3 The central tension

We can now state the devastating observation that drives the rest of this paper. The isotropic spectrum—the condition of Jensen equality $[\mathbf{J}]$ —is simultaneously the *optimal* representation (it minimizes all convex risk functionals, by the inequalities above) and the *degenerate* one (all information is destroyed, all inputs become indistinguishable). The optimum is the catastrophe. This is not a coincidence; it is the central tension of representation learning.

Every method for learning representations must navigate this tension. The energy wants to decrease (pushing toward collapse $[\mathbf{C}]$). The task wants the encoder to discriminate (pushing away from collapse). In quantum mechanics $[\mathbf{Q}]$, the tension is exponentially sharper: the maximally mixed state $\rho = I/d$ is the global attractor, and $d = 2^n$ makes the basin of attraction exponentially wide. The gradient $[\nabla]$ pointing toward collapse is exponentially stronger than any gradient pointing away. Geometrically $[\mathbf{G}]$, the curvature of the loss landscape vanishes exponentially, leaving an exponentially flat table. The M_2 operator $[\mathbf{M}]$ quantifies this: its spectral gap measures how far the circuit ensemble is from the Haar measure, and for expressive circuits, the gap is $O(1/d^3)$ —exponentially small.

The mathematical question is: can we formalize conditions under which the discriminative force wins?

1.4 The classical fix, and why it may not scale

Classical JEPA (LeCun 2022, Balestriero–LeCun 2025) prevents collapse by regularization. VIREG drives the covariance toward $\Sigma = I$ (non-degenerate). SIGReg drives the embedding distribution toward a standard Gaussian (maximally spread). Both work by *breaking Jensen equality*: penalizing flat spectra.

Theorem 1.5 (Regularization Beats Collapse). *For any training loss L_{task} with regularizer R and strength $\lambda > 0$: if $R(\text{collapsed}) \gg R(\text{structured})$, then the collapsed state is not the minimum of $L_{\text{task}} + \lambda R$.*

Lean: `RegularizationTransfer.regularization_prevents_barren_plateau`.

But when the representation space is $\mathbb{C}^{2^n}$, the covariance matrix has 4^n entries. The number of training points is polynomial. Does the regularizer have enough data to even detect collapse in an exponentially large space?

Worse: in $\mathbb{C}^{2^n}$, not just the loss but its *gradient* vanishes. The landscape is not merely flat at the minimum; it is flat everywhere $[\nabla]$. The concentration of kernel values $[\mathbf{M}]$ around the Haar floor $\beta = 1/d(d+1)$ is Jensen equality for the doubled Hilbert space $[\mathbf{J}]$. The noise contraction q^{2L+1} is a discrete Ricci flow driving toward zero curvature $[\mathbf{G}]$. Every theme converges on the same prediction: the quantum version is exponentially harder.

To see exactly how, we must enter the quantum setting.

2 Movement II: The Quantum Dilemma

2.1 From $\mathbb{R}^d$ to $\mathbb{C}^{2^n}$

An n -qubit system lives in $\mathcal{H} = \mathbb{C}^{2^n}$, dimension $d = 2^n$. A **quantum feature map** encodes classical data into a quantum state:

$$\phi(x) = U(x)|0\rangle^{\otimes n}, \quad \rho(x) = |\phi(x)\rangle\langle\phi(x)|.$$

The kernel is the squared fidelity:

$$K(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2 = \text{Tr}[\rho(x)\rho(y)].$$

Theorem 2.1 (QFM is a Frozen Encoder). *The density matrix map $x \mapsto \rho(x)$ is a classical frozen encoder in the Hilbert–Schmidt space. The kernel $K(x, y) = \text{Tr}[\rho(x)\rho(y)]$ is positive semidefinite.*

Lean: `QJEPa.qfm_is_frozen_encoder`, from `TensorPSD.mstate_kernel_psd`.

This is not an analogy. Quantum feature maps *are* kernel methods (Schuld 2021). The density matrices $\rho(x)$ are the feature vectors. The kernel $K(x, y) = \text{Tr}[\rho(x)\rho(y)]$ is PSD by the Gram matrix theorem applied in the Hilbert–Schmidt inner product space. Every theorem about kernel methods—including the spectral characterization of collapse [C], the Jensen bounds on spectral degeneracy [J], and the curvature interpretation [G]—transfers immediately to the quantum setting. The only difference is the dimension: $d = 2^n$ instead of d .

For $n = 50$ (a modest quantum computer), $d \approx 10^{15}$. The covariance matrix of a representation in this space has $d^2 \approx 10^{30}$ entries. No classical computer can store it, let alone regularize it. Yet the quantum circuit manipulates this space natively, one gate at a time. The question is whether this native access provides any advantage, or whether the exponential dimension defeats us.

The bridge between classical and quantum energies:

$$E_Q = 1 - |\langle \phi(x) | \phi(y) \rangle|^2 \leq 2(1 - \text{Re} \langle \phi(x) | \phi(y) \rangle) = E_{\text{cl}}.$$

Lean: `Bridge.bridge_inequality`.

So quantum energy is bounded by classical energy. Collapse in the classical sense (zero classical energy) implies collapse in the quantum sense (zero quantum energy). But the converse is stronger: quantum states can have $E_Q = 0$ (perfect fidelity) without having $E_{\text{cl}} = 0$ (they may differ by a global phase). This asymmetry is the first hint that the quantum setting is structurally different: the equivalence classes are larger, the collapse basin is wider, and the landscape is flatter.

All six themes revisited at greater depth. We now re-encounter every theme from Movement I, but with quantum content:

- J Jensen.** The Haar concentration floor $\beta = 1/(d(d+1))$ is Jensen equality for the kernel spectrum. The quantum version of collapse is spectral isotropy of the kernel Gram matrix—exactly the condition $f(\bar{\lambda}) = f(\bar{\lambda})$ for $f = x^2$ on the doubled space $\mathcal{H} \otimes \mathcal{H}$.
- C Collapse.** Quantum collapse is worse than classical: not just the loss value but its gradient vanishes. If $K(x, y) \approx \text{const}$ for all θ (concentration), then $\partial K / \partial \theta \approx 0$ everywhere. Collapse and barren plateaus are the same phenomenon at different levels.
- G Geometry.** The noise contraction q^{2L+1} per L layers is a *discrete Ricci flow*. Each layer contracts distances by factor q . The critical depth L^* is where the flow reaches the fixed point: the maximally mixed state, which has zero curvature.
- M M_2 .** The concentration bound comes from M_2 ’s spectral gap. For Haar circuits, M_2 has eigenvalue $1/(d(d+1))$ on Sym^2 and $1/(d(d-1))$ on Λ^2 . The gap between them governs the rate of concentration.
- ∇ Gradient.** Five faces of impossibility, all arising from the *same* spectral isotropy at different levels: concentration of values, vanishing of gradients, noise contraction, effective depth truncation, measurement cost explosion.

2.2 The concentration catastrophe

For sufficiently expressive circuits, collapse is not a risk—it is an inevitability:

Theorem 2.2 (Exponential Concentration (Thanasilp et al. 2024)). *For n -qubit quantum feature maps forming an approximate 2-design:*

$$\text{Var}_{x,y}[\kappa(x, y)] \leq \beta_n = \frac{1}{d(d+1)}, \quad d = 2^n.$$

All kernel values concentrate exponentially around $\mu = 1/(d+1)$.

Lean: `Concentration.haarBeta_pos, haarBeta_le_inv_sq`.

Geometrically [G]: this is volume collapse. In $\mathbb{C}^d$ for large d , the surface area of the unit sphere concentrates in an exponentially thin equatorial band. All inner products look the same. The metric $d(x, y)^2 = K(x, x) + K(y, y) - 2K(x, y)$ collapses to zero: all points become equidistant.

The gradient consequence is immediate:

Theorem 2.3 (Barren Plateau). *For the kernel training loss $L_{\text{task}}(\theta) = f(\kappa_\theta)$:*

$$\text{Var}_\theta \left[\frac{\partial L_{\text{task}}}{\partial \theta_k} \right] \leq O(1/4^n).$$

Training by gradient descent is impossible.

Why does value concentration [C] imply gradient vanishing [V]? Because $\partial K / \partial \theta$ measures how much the kernel changes under parameter perturbation. If $K(x, y) \approx 1/(d+1)$ for *all* θ , there is nothing to change. The landscape is not just flat at the minimum; it is flat *everywhere*. This is why barren plateaus are worse than local minima. A local minimum has steep walls nearby: a sufficiently large perturbation can escape. A barren plateau has no slopes to follow at all. The gradient does not point “uphill” or “downhill”—it points nowhere, because the function varies by less than $1/4^n$ across the entire parameter space.

The connection to Jensen [J] is exact: the kernel spectrum is isotropic (all eigenvalues equal to $1/d(d+1)$), so any convex functional achieves its Jensen bound. The gradient, being a derivative of such a functional, inherits the flatness. Concentration of values and vanishing of gradients are not two separate phenomena—they are two consequences of the same spectral isotropy. The M_2 operator [M] makes this precise: its near-zero spectral gap forces all derived quantities to concentrate. Geometrically [G], the curvature tensor of the parameter space is exponentially small, so no direction of improvement can be detected.

2.3 The five faces of impossibility

F1 Concentration: $\text{Var}[\kappa] \leq 1/(d(d+1))$ for expressive circuits.

F2 Noise: depolarizing noise contracts the kernel by q^{2L+1} per L layers.

F3 Effective depth: non-unitary noise truncates circuits to $O(\log n)$ effective layers.

F4 Dequantization: shallow circuits admit classical tensor-network surrogates.

F5 Measurement cost: distinguishing the kernel from random requires 4^n shots.

Lean: `DeepConnections.qml.dilemma_complete` bundles all five.

The five faces form a *closed loop* of impossibility. Start with an expressive circuit (deep, many parameters). F1 says its kernel concentrates [C]. Try to fix this by adding noise tolerance—but F2 says noise contracts the signal exponentially [G]. Accept the noise and use only shallow circuits—but F3 says effective depth is $O(\log n)$, and F4 says such shallow circuits are classically simulable [Q]. Accept the classical simulation and try to use the quantum device only for speed—but F5 says even verifying the kernel value requires 4^n measurements. Every escape from one face runs into another.

The closed loop has a precise formulation in terms of M_2 [M]. The spectral gap Δ of M_2 controls F1 (concentration $\sim 1/\Delta$). Depth contraction controls F2 ($\Delta(L) \sim q^{2L+1}\Delta(0)$). The critical depth L^* where $\Delta(L^*) \sim O(1)$ controls F3. Tensor factorization of M_2 controls F4. The measurement cost $\sim 1/\Delta^2$ controls F5. The loop is a loop in the spectral space of M_2 .

2.4 The phase transition

Definition 2.4 (Kernel SNR). $\text{SNR}(L, n, q) = q^{2L+1} \cdot d(d+1)$.

Theorem 2.5 (Critical Depth). *The SNR crosses 1 at*

$$L^* = \frac{n \ln 2}{|\ln q|} - \frac{1}{2}.$$

For $L < L^$, signal dominates. For $L > L^*$, concentration wins.*

Lean: `CriticalDepth.phase_transition_characterization`.

Geometrically $[\mathbf{G}]$: the noise is a discrete Ricci flow. Total contraction after L layers is q^{2L+1} . The critical depth L^* is where the flow reaches its fixed point: $\rho = I/d$ $[\mathbf{C}]$, purity $= 1/d$ $[\mathbf{J}]$, zero curvature $[\mathbf{G}]$, zero spectral gap of M_2 restricted to the surviving subspace $[\mathbf{M}]$, vanishing gradient $[\nabla]$. All six themes diagnose the same phase transition.

Lean: `GeometricUnification.ricci_flow_converges`, `ricci_flow_monotone`.

The five faces are not independent obstacles—they are five theorems about the same spectral condition. Through the lens of M_2 $[\mathbf{M}]$: F1 is the statement that M_2 ’s spectral gap vanishes for 2-designs. F2 is the statement that depth contracts the gap: $\Delta(L) \sim q^{2L+1}\Delta(0)$. F3 says the contraction reaches the fixed point at L^* . F4 says that when M_2 factorizes as a tensor product, the circuit is classically simulable. F5 says that the measurement cost to distinguish M_2 from the Haar value scales as $1/\Delta^2 = O(4^n)$. One operator, five faces.

Through the lens of Jensen $[\mathbf{J}]$: all five faces are consequences of spectral isotropy. F1 is Jensen equality for the kernel spectrum. F2 is a contraction toward Jensen equality (noise drives spectra toward flatness). F3 is the time at which Jensen equality is reached. F4 says that Jensen equality in the doubled space implies classical simulability. F5 is the sample complexity of detecting deviation from Jensen equality.

All five obstructions arise from *global* properties: the full 2^n -dimensional state, the global kernel, total depth, global measurement statistics. What if we restrict attention to *local* properties? A local observable sees only a polynomial-dimensional marginal. Could locality escape the global curse?

3 Movement III: The Solution—Locality and Curvature

3.1 The gradient variance identity

We need more than showing the collapsed state is suboptimal (Theorem 1.5). We need the gradient to be non-vanishing. This requires a new theorem.

Theorem 3.1 (Gradient Variance Identity). *For a regularized loss $L_{\text{reg}} = L_{\text{task}} + \lambda R$, let $\varepsilon = \text{Var}[\partial L_{\text{task}}/\partial \theta]$ and $\sigma^2 = \text{Var}[\partial R/\partial \theta]$. Then:*

$$\text{Var}\left[\frac{\partial L_{\text{reg}}}{\partial \theta}\right] \geq (\sqrt{\varepsilon} - \lambda\sigma)^2.$$

Proof. By variance decomposition: $\text{Var}[A + \lambda B] = \text{Var}[A] + \lambda^2 \text{Var}[B] + 2\lambda \text{Cov}[A, B]$. By Cauchy–Schwarz: $|\text{Cov}[A, B]| \leq \sqrt{\text{Var}[A] \cdot \text{Var}[B]}$. Therefore $\text{Var}[A + \lambda B] \geq \varepsilon + \lambda^2 \sigma^2 - 2\lambda \sigma \sqrt{\varepsilon} = (\sqrt{\varepsilon} - \lambda\sigma)^2$. $\square$

Lean: `GradientBound.gradient_variance_identity`.

The identity says: the regularized gradient variance is a perfect square, hence always non-negative. More importantly, **it is strictly positive whenever $\sqrt{\varepsilon} \neq \lambda\sigma$** . In the barren plateau regime $[\mathbf{C}]$, $\varepsilon \approx 0$ (exponentially small), so the condition becomes $0 \neq \lambda\sigma$, which holds whenever $\sigma > 0$ and $\lambda > 0$. The entire burden of the proof now shifts to showing $\sigma > 0$: the regularizer’s gradient must not itself concentrate.

All six themes at full depth. The gradient identity connects to every theme:

- J Jensen.** The identity $(\sqrt{\varepsilon} - \lambda\sigma)^2 \geq 0$ is *itself* a Jensen-type inequality: the variance of a sum is bounded by the square of the difference of standard deviations. This is Cauchy–Schwarz, which is Jensen for $f = x^2$ on the covariance inner product. Jensen appears at every level of the argument.
- C Collapse.** The purity regularizer $R = \sum_k \text{Tr}[\rho_k^2]$ penalizes the *same* isotropy that Jensen identifies. Purity $= 1/d$ (maximally mixed = collapsed) iff Jensen equality for $f = x^2$. The regularizer *breaks* Jensen equality.
- Q Quantum.** Locality is the escape: a local observable sees only a polynomial-sized light cone. The global-vs-local distinction is the quantum analog of the bias-variance tradeoff—restricting the observable class reduces variance (concentration) at the cost of information (bias).
- G Geometry.** The fibre bundle is now explicit. The encoder ϕ is a section of the bundle $(X, \mathcal{H}, U(d))$. The predictor V is the connection. The energy *is* the curvature. The Lie algebra dimension determines trainability, simulability, and curvature simultaneously.
- M M_2 .** The local purity gradient bound $\sigma^2 \geq c$ comes from M_2 restricted to local operators having spectral gap $\geq c$.
- ∇ Gradient.** The gradient bound is a curvature bound: the regularizer adds curvature $\lambda^2 c/4$ to the loss landscape, ensuring the landscape has slope everywhere.

3.2 Local purity regularization

Theorem 3.1 tells us: if the regularizer’s gradient does not concentrate, the total gradient is bounded away from zero. But the regularizer must be chosen carefully. A *global* regularizer—one depending on the full $2^n \times 2^n$ density matrix—would itself suffer from barren plateaus (Cerezo et al. 2021). Its gradient would vanish as $O(1/4^n)$, and we would merely have moved the barren plateau from L_{task} to R . The regularizer must be **local**.

The purity of a quantum state is $\text{Tr}[\rho^2] = \sum_i \lambda_i^2$. By QM-AM (Jensen with $f = x^2$) [J]: $\text{Tr}[\rho^2] \geq 1/d$, with equality iff $\rho = I/d$ (maximally mixed = collapsed [C]).

The purity regularizer is a sum of *local* purities:

$$R(\theta) = \sum_{k=1}^n \text{Tr}[\rho_k(\theta)^2], \quad \rho_k = \text{Tr}_{-k}[\rho].$$

Each term depends on the 1-qubit reduced state ρ_k —a **local** observable [Q]. Its gradient does not concentrate:

Theorem 3.2 (Local Purity Bound). *For the purity regularizer, $\sigma^2 \geq c > 0$ for a universal constant c independent of n .*

Status: Axiomatized (`local_cost_gradient_bound`), justified by numerical evidence:

n	d	$\text{Var}[\nabla L_{\text{global}}]$	$\text{Var}[\nabla L_{\text{local}}]$	Ratio
4	16	4.1×10^{-3}	1.7×10^{-2}	$4\times$
6	64	4.3×10^{-4}	1.6×10^{-2}	$38\times$
8	256	3.2×10^{-4}	1.9×10^{-2}	$59\times$
10	1024	$\sim 10^{-5}$	1.9×10^{-2}	$\sim 10^3\times$

The global gradient variance decays exponentially ($\sim 0.36^n$). The local gradient variance is **flat at** ≈ 0.017 , independent of n . This is the locality principle in action: a local observable [Q] sees only a polynomial-sized light cone [G], so its gradient [∇] does not inherit the exponential concentration [M] that Jensen [J] predicts for global observables in the collapsed regime [C].

Corollary 3.3 (No Barren Plateau). *For purity-regularized Q-JEPA with $\lambda > 0$:*

$$\text{Var}\left[\frac{\partial L_{\text{reg}}}{\partial \theta}\right] \geq \frac{\lambda^2 c}{4} > 0.$$

The gradient does not vanish. Training is possible.

Lean: `FullChain.no_barren_plateau_quantitative`.

3.3 The fibre bundle

The Q-JEPA framework $(X, \mathcal{H}, U(d))$ is a **fibre bundle** $[\mathbf{G}]$:

- **Base space** $B = X$: the input data manifold.
- **Fibre** $F = \mathcal{H} = \mathbb{C}^{2^n}$: the quantum Hilbert space.
- **Structure group** $G = U(d)$: the unitaries.
- **Section** $s(x) = \phi(x)$: the encoder.
- **Connection** $\nabla = V$: the predictor (parallel transport).

Theorem 3.4 (Curvature = Energy). *The Q-JEPA energy $E(x, y) = \|\phi(y) - V(\phi(x))\|^2$ is the holonomy defect of the connection V .*

- *Zero curvature ($E = 0$ for all pairs) $\Leftrightarrow$ collapsed section or perfect predictor $[\mathbf{C}]$.*
- *Positive curvature ($E > 0$ somewhere) $\Leftrightarrow$ non-trivial geometry (encoder discriminates).*

Lean: `Geometric.curvature_eq_energy, zero_curvature_iff, collapsed_flat_zero`.

The fibre bundle perspective explains why the same energy functional governs both learning and geometry. In differential geometry, the curvature of a connection measures the failure of parallel transport to be path-independent. In Q-JEPA, the predictor V is parallel transport, and the energy measures the failure of prediction. A “flat” connection (zero curvature) means parallel transport is trivial—the predictor perfectly reconstructs the encoder at every point. But a trivial predictor means the encoder carries no information that varies across inputs. **Flatness is collapse** $[\mathbf{C}]$. Conversely, positive curvature means the connection is non-trivial—the predictor must do real work. The amount of work (the energy) is the curvature. So **learning = curvature, and collapse = flatness**, and this is not a metaphor but an identification of mathematical structures.

The fibre bundle is not an afterthought—it is the *reason* locality works. A local observable $[\mathbf{Q}]$ probes a polynomial-dimensional subbundle. The curvature $[\mathbf{G}]$ of the subbundle need not vanish even when the global bundle’s curvature concentrates. This is because the restriction to a subbundle corresponds to restricting M_2 $[\mathbf{M}]$ to local operators, and on this restricted subspace, M_2 retains a spectral gap because the polynomial-dimensional invariant subspace is not large enough for concentration to take hold. The gradient bound $[\nabla]$ is a curvature bound: the regularizer adds curvature $\lambda^2 c/4$ to the loss landscape, where c comes from M_2 ’s local spectral gap. Jensen $[\mathbf{J}]$ governs the global spectrum but does *not* force isotropy on the local restriction—the local probe sees a non-degenerate slice of the spectrum.

3.4 The Lie algebra trichotomy

The gates of the circuit generate a Lie algebra $\mathfrak{g} \subseteq \mathfrak{u}(d)$ whose dimension determines everything:

Theorem 3.5 (Lie Algebra Trichotomy). *For a parameterized quantum circuit with dynamical Lie algebra $\mathfrak{g}$:*

- $\dim(\mathfrak{g}) = \text{poly}(n) \Rightarrow \text{trainable } [\nabla] + \text{classically simulable } [\mathbf{Q}]$.
- $\dim(\mathfrak{g}) = \exp(n) \Rightarrow \text{barren plateau } [\nabla] + \text{possibly advantageous } [\mathbf{Q}]$.
- $\dim(\mathfrak{g}) = \exp(n) + \text{noise} \Rightarrow \text{barren plateau} + O(\log n) \text{ depth} + \text{simulable}$.

Lean: `LieAlgebraDim.poly_lie_trainable, exp_lie_concentrated, equivariance_restricts`.

The dimension $\dim(\mathfrak{g})$ is the order parameter. It determines which regime we occupy: trainable but simulable $[\mathbf{Q}]$, or expressive but flat $[\mathbf{G}]$. The sharpness is absolute—there is no intermediate regime. G -equivariance restricts $\mathfrak{g}$ to the centralizer of G , via Schur–Weyl duality $[\mathbf{J}]$: $V \otimes V = \text{Sym}^2(V) \oplus \Lambda^2(V)$ decomposes the doubled space into the same invariant subspaces that M_2 $[\mathbf{M}]$ diagonalizes.

Lean: `SchurWeyl.involution_sym_idempotent, sym_anti_complementary`.

The regularization strategy of Movement I does not change the Lie algebra—it changes the loss landscape *within a fixed Lie algebra*, adding curvature $[\mathbf{G}]$ where there was none. This is the sense in which the gradient bound $[\nabla]$ and the collapse criterion $[\mathbf{C}]$ operate at different levels: the Lie algebra determines what is *reachable*; the regularizer determines what is *findable*.

This distinction is crucial. One cannot “partially” avoid barren plateaus by making the Lie algebra “medium-sized.” Either the algebra is polynomial and the circuit is trainable but simulable, or the algebra is exponential and the circuit is expressive but untrainable $[\mathbf{Q}]$. The regularizer does not change this dichotomy—it operates *within* one regime, ensuring that the gradient $[\nabla]$ remains detectable even when the global M_2 spectrum $[\mathbf{M}]$ has concentrated. The local spectral gap of M_2 is the bridge: it lives within the polynomial-dimensional subalgebra of local operators, where Jensen $[\mathbf{J}]$ does not force isotropy, and the curvature $[\mathbf{G}]$ of the local subbundle remains positive.

4 Movement IV: The Universal Object

4.1 The second moment operator

We have seen that concentration, trainability, simulability, and curvature are governed by the same structure. We now identify the single object that unifies them.

Definition 4.1 (Second Moment Operator). *For a parameterized circuit ensemble $\{U(\theta)\}_\theta$, the **second moment operator** is*

$$M_2 = \mathbb{E}_\theta [U(\theta)^{\otimes 2} \otimes U(\theta)^{* \otimes 2}],$$

a positive semidefinite operator on $\mathcal{H} \otimes \mathcal{H}$.

The doubled Hilbert space decomposes via Schur–Weyl:

$$\mathcal{H} \otimes \mathcal{H} = \text{Sym}^2(\mathcal{H}) \oplus \Lambda^2(\mathcal{H}).$$

Theorem 4.2 (Haar M_2 Spectrum). *For Haar-random circuits, M_2 has eigenvalues:*

- On $\text{Sym}^2(\mathcal{H})$ (dimension $d(d+1)/2$): eigenvalue $\lambda_{\text{sym}} = \frac{1}{d(d+1)}$.
- On $\Lambda^2(\mathcal{H})$ (dimension $d(d-1)/2$): eigenvalue $\lambda_{\text{anti}} = \frac{1}{d(d-1)}$.

The spectral gap is $\Delta = \lambda_{\text{anti}} - \lambda_{\text{sym}} = \frac{2}{d(d^2-1)}$. The trace $\text{Tr}[M_2] = 1$.

Lean: `MomentOperator.haarM2, haar_spectral_gap, haar_m2_trace_one`.

All six themes at maximum depth, fully unified.

J Jensen. M_2 ’s spectral decomposition on $\text{Sym}^2 \oplus \Lambda^2$ is the Schur–Weyl decomposition, which is the representation-theoretic content of Jensen’s inequality applied to the doubled Hilbert space. The two eigenvalues λ_{sym} and λ_{anti} are the Jensen bounds for symmetric and antisymmetric probes. When the gap vanishes ($\lambda_{\text{sym}} = \lambda_{\text{anti}}$), Jensen equality holds on $\mathcal{H}^{\otimes 2}$ —this is the representation-theoretic *meaning* of collapse.

C Collapse. Collapse = M_2 having zero spectral gap. The concentrated state $\rho = I/d$ is the fixed point of the M_2 channel: applying the M_2 averaging to any state drives it toward the maximally mixed state, just as iterating Jensen equality drives any spectrum toward isotropy.

Q Quantum. During numerical computation of M_2 , we discovered that the commonly used Ry+CZ circuit ansatz generates only **real orthogonal** unitaries $O(d)$, not the full unitary group $U(d)$. This is because R_y and CZ gates both have real matrix entries; compositions of real matrices stay real. Consequence: M_2 for Ry+CZ saturates at Frobenius distance ≈ 0.16 from Haar- $U(d)$, with leading eigenvalue $1/d = 0.25$ instead of converging to $1/(d(d+1)) \approx 0.05$, even at infinite depth. Circuit structure constrains M_2 . To reach $U(d)$, a gate with complex entries (such as R_z) is required. This discovery illustrates a key point: the M_2 operator is not just a theoretical construct but a *diagnostic tool* that reveals structural limitations invisible to other methods.

G Geometry. M_2 is the curvature tensor of the parameter-averaged bundle on $\mathcal{H}^{\otimes 2}$. Its spectral gap is the sectional curvature: a large gap means high curvature (rich geometry, trainable), a small gap means flat (collapsed, untrainable). The holonomy representation has characters equal to M_2 's eigenvalues. Depth contracts the gap: $\Delta(L) \sim q^{2L+1} \cdot \Delta(0)$, the same Ricci flow as Theorem 2.5.

M M_2 determines everything:

Property	What M_2 gives	Lean reference
Concentration	Spectral gap of M_2	<code>SpectralM2.spectral_gap_bounds_variance</code>
Trainability	M_2 restricted to local ops	<code>SpectralM2.local_eigenvalue_gives_gradient_bound</code>
Simulability	M_2 factorizes as tensor	(connections to <code>Matchgate.lean</code>)
Curvature	$M_2 =$ averaged curvature	<code>GeometricM2.depth_contracts_gap</code>
Holonomy	$\dim(\mathfrak{g}) = \text{rank}(M_2)$	<code>LieAlgebraDim.equivariance_restricts</code>

∇ Gradient. The universal no-barren-plateau theorem follows from M_2 's local spectral gap alone.

Verified: `tools/sage/moment_operator.py`, `tools/sage/rg_flow.py`.

This is why the same pattern kept recurring throughout the paper. Collapse (Movement I), concentration (Movement II), trainability (Movement III), curvature (Movement III) — they all felt like aspects of a single phenomenon because they *are*. They are different projections of the M_2 spectrum onto different subspaces. Jensen's inequality [J] governs M_2 just as it governs any PSD operator: spectral flatness of M_2 is the universal obstruction, and spectral gap in M_2 is the universal remedy. The central tension of the paper—the optimum is the catastrophe—is the statement that M_2 's spectral gap vanishes for maximally expressive circuits. The resolution—local purity regularization—is the statement that the *local* spectral gap of M_2 does not vanish, even when the global gap does.

M_2 is to quantum representation learning what the Riemann curvature tensor is to differential geometry: the single object from which all local geometric information can be extracted. The analogy is not loose: M_2 is an averaged curvature tensor on $\mathcal{H}^{\otimes 2}$, as Theorem 3.4 makes precise when lifted to the doubled space.

4.2 The universal theorem

Theorem 4.3 (Universal No-Barren-Plateau). *For any circuit family whose M_2 has spectral gap $\geq c > 0$ on local operators, purity regularization with $\lambda > 0$ gives:*

$$\text{Var} \left[\frac{\partial L_{\text{reg}}}{\partial \theta} \right] \geq (\sqrt{\varepsilon} - \lambda\sqrt{c})^2 > 0,$$

when ε (task gradient variance) is small enough that $\sqrt{\varepsilon} < \lambda\sqrt{c}$.

Lean: `UniversalM2.universal_no_barren_plateau`.

The power of Theorem 4.3 is its universality. Previous no-barren-plateau results required specific architectures and specific loss functions. This theorem requires only a spectral condition on M_2 —a single number computable from the circuit architecture alone, without reference to training data or loss function. The M_2 operator is a property of the *hypothesis class*, not the *learning problem*.

The proof chain recapitulates all six themes:

1. The JEPA energy defines the learning problem [C].
2. Jensen’s inequality characterizes collapse as spectral isotropy [J].
3. The quantum feature map embeds the problem in $\mathbb{C}^{2^n}$ [Q].
4. The fibre bundle identifies energy with curvature [G].
5. The gradient identity bounds the regularized gradient $[\nabla]$.
6. M_2 ’s local spectral gap supplies the bound $c > 0$ [M].

Every link is a theorem in Lean. The six themes are not six perspectives on one result—they are six necessary steps in its proof.

4.3 What we have proved

Let us take stock. The four movements have established:

1. **Jensen on spectra** [J] governs both optimal representations and collapse. The isotropic spectrum is simultaneously the best (minimizes all convex risk functionals) and the worst (destroys all information). The optimum is the catastrophe.
2. **Collapse equivalence** [C]: constant encoder $\Leftrightarrow$ zero energy $\Leftrightarrow$ isotropic spectrum $\Leftrightarrow$ Jensen equality for $f = x^2$.
3. **Quantum feature maps are kernel methods** [Q]: $\rho(x)$ is a frozen encoder, $K = \text{Tr}[\rho(x)\rho(y)]$ is PSD, and the bridge inequality relates classical and quantum energies.
4. **Five faces of the QML dilemma**: concentration [C], noise [G], effective depth, de-quantization [Q], measurement cost—all arising from the same spectral isotropy.
5. **Critical depth** [G]: $L^* = n \ln 2 / |\ln q| - 1/2$, the Ricci flow fixed point.
6. **Gradient variance identity** $[\nabla]$: $\text{Var}[\nabla L_{\text{reg}}] \geq (\sqrt{\varepsilon} - \lambda\sigma)^2$, a perfect square.
7. **Local purity prevents barren plateaus**: $\sigma^2 \geq c > 0$ uniformly in n , because local observables see polynomial-sized light cones.
8. **Curvature = energy** [G]: the fibre bundle $(X, \mathcal{H}, U(d))$ with holonomy defect = Q-JEPA energy.
9. M_2 **unifies everything** [M]: spectral gap = concentration rate, rank = $\dim(\mathfrak{g})$, local eigenvalue = gradient bound. One operator, one spectrum, five consequences.

All of the above is formalized in Lean 4 with 0 sorry, verified across 6 independent tools. The chain of reasoning visits all six themes at every step—not because we chose to organize it this way, but because the mathematics demands it.

4.4 The open question

Question 4.4. *Does there exist a circuit family $\{C_n\}$ such that:*

- (a) M_2 has spectral gap $\geq c > 0$ on local operators (trainable) $[\nabla]$.
- (b) No classical algorithm computes $K(x, y)$ in $\text{poly}(n)$ time (non-simulable) [Q].
- (c) The purity is maintainable via SWAP test feedback (physically realizable) [M].

The tension is between (a) and (b). The Lie algebra trichotomy [G] (Theorem 3.5) suggests they may be incompatible: $\dim(\mathfrak{g}) = \text{poly}$ gives (a) but not (b); $\dim(\mathfrak{g}) = \text{exp}$ gives (b) but not (a). Jensen [J] makes the tension precise: the spectral gap on local operators (a) is a Jensen-type lower bound, while classical non-simulability (b) requires the *global* spectrum to be sufficiently non-isotropic. Whether local non-isotropy can coexist with global non-simulability is the core mathematical question.

The escape route may be *mid-circuit measurement and feedback*, which breaks the Lie algebra framework. Measurement is not a unitary operation—it collapses the state [C], projecting onto a subspace. The post-measurement state lives in a smaller Hilbert space, and the Lie algebra of the remaining circuit acts on this reduced space. If the measurement is chosen adaptively (based on previous outcomes), the effective Lie algebra changes at each step. This means the trichotomy—which assumes a fixed Lie algebra for the entire circuit—no longer applies.

The circuit can be “locally polynomial” (trainable at each step, with polynomial Lie algebra [G]) while “globally non-simulable” (the adaptive sequence of Lie algebras cannot be efficiently reproduced classically [Q]). Each measurement step maintains a local spectral gap in M_2 [M], preserving trainability [V], while the composition of adaptive measurements builds correlations that no polynomial-time classical algorithm can reproduce. Jensen [J] guarantees local non-isotropy at each step, but the global spectral structure across the adaptive sequence may be non-isotropic in a way that is computationally hard to simulate.

Whether this escape route actually works—whether adaptive circuits can achieve all three conditions simultaneously—is the central open problem.

The tools developed here— M_2 [M], the spectral gap framework [J], the gradient identity [V], the fibre bundle [G]—provide the language in which the question can be precisely stated and, we hope, eventually resolved. The resolution of this question determines whether quantum advantage in representation learning is possible.

The fugue resolves. We began with an energy and its pathology—a trivial minimizer that collapses all representations. We showed that the pathology is governed by a single principle (Jensen [J]) that recurs at every level: in the kernel spectrum [C], in the quantum concentration bound [Q], in the curvature of the fibre bundle [G], in the gradient variance [V], and ultimately in the spectral gap of M_2 [M]. The solution—local purity regularization—works because it breaks Jensen equality locally, adding curvature to a globally flat landscape. The six themes are not six stories about one result. They are six voices in a single fugue, each entering in its own register, each carrying the same subject: *spectral isotropy is both the destination and the disease, and the cure is local non-isotropy*.

5 Verification

Tool	What it verifies	Status
Lean 4 + Mathlib	572 declarations, 0 <code>sorry</code> , 4 axioms	Kernel-verified
SageMath 10.8	M_2 computation, Haar axiom, scaling data	Exact + numerical
Isabelle 2025	Spectral gap, trace normalization, gradient chain	All pass
Coq/Rocq 9.1	Gradient bound chain, local purity chain	All compile
Agda 2.7	Structural content: locality, decomposition	Type-checked
NumPy 2.4	Gradient scaling $n = 4$ –10, M_2 eigenvalues	Monte Carlo

The 4 axioms:

1. `haar_fourth_moment`: Haar integral = Weingarten contraction (representation theory).
2. `local_cost_gradient_bound`: local purity gradient $\sigma^2 \geq c > 0$ (light-cone argument).
3. `poly_lie_trainable`: $\dim(\mathfrak{g}) = \text{poly} \Rightarrow \text{trainable}$ (Ragone/Cerezo 2024).
4. `m2_exists_with_spectrum`: M_2 spectral gap bounds are uniform (unified axiom).

Each axiom is justified by cross-verification in at least 5 of the 6 tools.

Acknowledgments

Formalization uses Mathlib v4.29.1 and Physlib (leanprover-community). Papers synthesized: Balestrieri–LeCun 2025, Thanasilp et al. 2024, Schuld 2021, LeCun 2022, Assran et al. 2023, Bardes 2022, Havlicek et al. 2019, Jozsa–Miyake 2008, Collins–Śniady 2006, Valiant 2002, Mele et al. 2026, Landman et al. 2024, Ragone–Cerezo 2024, Cerezo 2021, Larocca et al. 2024.